

A NON-STANDARD $(5, 4)$ BROADCAST OF THE INFINITE GRID OF DENSITY $1/20$

ABSTRACT. Blessing, Insko, Johnson, and Mauretour conjectured that the optimal (t, r) and $(t - 1, r - 2)$ broadcast dominating sets of $\mathbb{Z} \times \mathbb{Z}$ have equal density, equivalently that $\delta_{t,r} = \delta_{t+1,r+2}$ for all $r \leq t$. That conjecture was already known to fail at other parameters; we add the instance $(t, r) = (4, 2)$, that is the pair of cells $(4, 2)$ and $(5, 4)$. The sublattice $S = \langle (10, 0), (4, 2) \rangle$ of index 20, obtained by doubling the perfect Lee code of radius one, is a $(5, 4)$ broadcast dominating set of $\mathbb{Z}^2$, so

$$\delta_{5,4} \leq \frac{1}{20} < \frac{1}{18} = \delta_{4,2}.$$

The certificate is the table of twenty coset receptions. S is not a standard broadcast: it has no vertex on any odd horizontal line, and $\mathbb{Z}^2/S \cong \mathbb{Z}/2 \oplus \mathbb{Z}/10$ is not cyclic. We do not determine $\delta_{5,4}$; with a quoted lower bound of Shlomi it lies in $\frac{1}{21} \leq \delta_{5,4} \leq \frac{1}{20}$.

1. THE CONJECTURE

Throughout, $\mathbb{Z}^2 = \mathbb{Z} \times \mathbb{Z}$ is the infinite grid graph, whose graph distance d is the ℓ_1 distance. Fix integers $1 \leq r \leq t$. A vertex v of a set $S \subseteq \mathbb{Z}^2$ — a *tower* — provides a vertex u with the signal $\max(0, t - d(u, v))$, so that a tower provides *itself* with t . The total signal received at u is

$$\text{rec}_S(u) = \sum_{v \in S} \max(0, t - d(u, v)),$$

the sum including the term $v = u$ when $u \in S$. The set S is a (t, r) *broadcast dominating set* if $\text{rec}_S(u) \geq r$ for *every* $u \in \mathbb{Z}^2$, towers included. Its density is $\delta(S) = \lim_{n \rightarrow \infty} |S \cap [-n, n]^2| / (2n+1)^2$, and $\delta_{t,r} = \delta_{t,r}(\mathbb{Z}^2)$ denotes the least density of a (t, r) broadcast dominating set of $\mathbb{Z}^2$. A *standard* broadcast is a lattice

$$T(d, e) = \langle (d, 0), (e, 1) \rangle = \{(x, y) \in \mathbb{Z}^2 : x \equiv ey \pmod{d}\},$$

one tower out of every d on *every* horizontal line; its density is $1/d$.

Blessing, Insko, Johnson, and Mauretour [1] close their paper with a list of conjectures and open problems, whose fourth item reads, verbatim:

“We have seen that $(3, 3)$ and $(2, 1)$ broadcast dominating sets are equal for large grids. We conjecture that the optimal (t, r) and $(t - 1, r - 2)$ broadcast dominating sets of $\mathbb{Z} \times \mathbb{Z}$ are equal. If this is true, then all (t, r) broadcast domination

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problems for large grids can be reduced to $(t, 2)$ and $(t, 1)$ broadcast domination problems.”

The parameter range $1 \leq r \leq t$ is fixed earlier in that paper, and the statement is made for the infinite grid, not for finite grids. Drews, Harris, and Randolph restate it as a conjecture [2]: “*For all $t, r \in \mathbb{Z}^+$, the optimal (t, r) and $(t + 1, r + 2)$ broadcast densities are equal.*” In the notation above the assertion is

$$(1) \quad \delta_{t,r} = \delta_{t+1,r+2} \quad \text{for all } 1 \leq r \leq t.$$

We refute (1) at $(t, r) = (4, 2)$.

2. THE WITNESS

Theorem 1. *Let*

$$S = \langle (10, 0), (4, 2) \rangle = \{(x, y) \in \mathbb{Z}^2 : y \equiv 0 \pmod{2}, x \equiv 2y \pmod{10}\}.$$

Then S is a $(5, 4)$ broadcast dominating set of $\mathbb{Z}^2$ of density exactly $1/20$. Consequently

$$\delta_{5,4} \leq \frac{1}{20} < \frac{1}{18} = \delta_{4,2},$$

so (1) is false at $(t, r) = (4, 2)$.

Towers sit on even horizontal lines only; within an occupied line they are 10 apart; each occupied line is shifted 4 to the right of the occupied line two below it. The two descriptions agree: $(x, y) = m(10, 0) + n(4, 2)$ has $y = 2n$ even and $x = 10m + 4n \equiv 2y \pmod{10}$, and conversely $y = 2n$ together with $x \equiv 2y \pmod{10}$ gives $x - 4n \equiv 0 \pmod{10}$. The basis matrix $\begin{pmatrix} 10 & 0 \\ 4 & 2 \end{pmatrix}$ has determinant 20, so S has index 20 in $\mathbb{Z}^2$ and the twenty classes of $\mathbb{Z}^2/S$ are represented by the pairs (x, y) with $0 \leq x \leq 9$ and $y \in \{0, 1\}$.

The certificate is the list of receptions at those twenty representatives:

	$x = 0, 1, \dots, 9$										sum
$y = 0$	5	4	5	4	5	4	5	4	5	4	45
$y = 1$	4	4	4	4	4	4	4	4	4	4	40
											85

The minimum is $4 = r$, attained at fifteen of the twenty classes. Two entries worked out by hand: the shortest nonzero vector of S is $(4, 2)$, of ℓ_1 norm $6 > 4$, so a tower receives nothing but its own 5, giving the entry at $(0, 0)$; and $(1, 0)$ is at distance 1 from the tower $(0, 0)$ and at distance ≥ 5 from every other tower — the next nearest are $(4, 2)$ and $(2, -4)$, both at distance 5 — so it receives exactly 4, which is the tight case.

The sum of the twenty numbers is forced, which checks the table as a whole.

Lemma 2. *Let $\Lambda \subseteq \mathbb{Z}^2$ be a sublattice of index D , let $t \geq 1$, and let R be any set of coset representatives of $\mathbb{Z}^2/\Lambda$. Then*

$$\sum_{u \in R} \text{rec}_\Lambda(u) = W_t := \sum_{z \in \mathbb{Z}^2} \max(0, t - \|z\|_1) = t + \sum_{m=1}^{t-1} 4m(t - m),$$

independently of Λ and of D . In particular $W_5 = 5 + 16 + 24 + 24 + 16 = 85$.

Proof. Write $f(z) = \max(0, t - \|z\|_1)$, a nonnegative function of finite support, so that $\text{rec}_\Lambda(u) = \sum_{\lambda \in \Lambda} f(u - \lambda)$. As u runs over R and λ over Λ , the difference $u - \lambda$ takes each value of $\mathbb{Z}^2$ exactly once, because R meets every coset of Λ exactly once. Hence $\sum_{u \in R} \text{rec}_\Lambda(u) = \sum_{z \in \mathbb{Z}^2} f(z) = W_t$. Counting cells by ℓ_1 norm — one of norm 0 and $4m$ of norm $m \geq 1$ — gives the displayed formula. $\square$

Proof of Theorem 1. The twenty entries of the table sum to $45 + 40 = 85 = W_5$, in agreement with Lemma 2. That agreement checks the table jointly and does not by itself bound the individual entries; read one by one, their minimum is 4. Since rec_S is invariant under translation by S , and the twenty representatives meet every class of $\mathbb{Z}^2/S$, we get $\text{rec}_S(u) \geq 4$ for every $u \in \mathbb{Z}^2$: the set S is a (5, 4) broadcast dominating set. It contains exactly one point per fundamental domain, of area 20, so $\delta(S) = 1/20$.

For the consequence, $\delta_{4,2} = 1/18$: Drews, Harris, and Randolph prove $\delta_{t,2} = 1/(2(t-1)^2)$ for all $t \geq 2$ [2], which at $t = 4$ gives $1/(2 \cdot 3^2) = 1/18$. Since $1/20 < 1/18$ and $\delta_{5,4} \leq \delta(S) = 1/20$, the two densities $\delta_{4,2}$ and $\delta_{5,4}$ differ, contradicting (1) at $(t, r) = (4, 2)$. $\square$

Remark (S as a doubled perfect Lee code). Let $\varphi(X, Y) = (2X, 2Y)$ and let $L = T(5, 2) = \langle (5, 0), (2, 1) \rangle = \{(X, Y) : X \equiv 2Y \pmod{5}\}$, the perfect Lee code of radius 1 [6] — equivalently the optimal (2, 1) broadcast, of density $1/5$: every cell of $\mathbb{Z}^2$ is within distance 1 of exactly one codeword. Then $S = \varphi(L)$ and $\delta(S) = \frac{1}{4} \cdot \frac{1}{5} = \frac{1}{20}$. Doubling doubles all ℓ_1 distances, so at $t = 5$ a tower $\varphi(v)$ reaches $\varphi(u)$ only when $d(u, v) \leq 2$, with signal 5, 3, 1 according as $d(u, v) = 0, 1, 2$. On $2\mathbb{Z}^2$ this settles the reception outright: a codeword receives its own 5 and nothing else, because the shortest nonzero vector of L has norm 3; a non-codeword receives 3 from the unique codeword at distance 1 together with 1 from each of the two codewords at distance 2, again 5. The receptions off $2\mathbb{Z}^2$ are not settled by this argument; the table above gives each of them as exactly 4.

Proposition 3. *S is not a standard broadcast, nor a translate of one, nor the image of one under any change of basis of $\mathbb{Z}^2$.*

Proof. Every point of S has even second coordinate, so S misses the horizontal line $y = 1$ entirely, whereas every $T(d, e)$ contains a point on every horizontal line; the same holds after translating. For the stronger statement, the basis $\{(d, 0), (e, 1)\}$ of $T(d, e)$ has an entry equal to 1, so the greatest common divisor of the entries of its basis matrix is 1, the invariant factors are $(1, d)$, and $\mathbb{Z}^2/T(d, e) \cong \mathbb{Z}/d$ is cyclic. All four entries of the basis matrix of S are even, so its invariant factors are $(2, 10)$ and $\mathbb{Z}^2/S \cong \mathbb{Z}/2 \oplus \mathbb{Z}/10$ is not cyclic. The isomorphism type of $\mathbb{Z}^2/\Lambda$ is invariant under $GL_2(\mathbb{Z})$, so no reflection, transposition, or relabelling of a standard broadcast is S . $\square$

3. SCOPE

What is refuted is the equality (1) at the single instance $(t, r) = (4, 2)$. Each instance of (1) is a separate statement and needs its own witness; we exhibit one only at $(4, 2)$. The equality was already refuted in print at other parameters: [3] reports counterexamples, and [4] restates that Drews, Harris, and Randolph “showed . . . that, in fact, $\delta_{t+1, r+2}(\mathbb{Z}^2) < \delta_{t, r}(\mathbb{Z}^2)$ for several values of t and r ”. The instance $(4, 2)$ falls in the sub-case $r = 2$ with t even and $t > 3$ singled out in [3].

We do not determine $\delta_{5,4}$. Shlomi [5] proves $\delta_{t,r} \geq r/C_{t,r}$ with $C_{t,r} = \frac{2}{3}r^3 - 2r^2t + 2rt^2 + \frac{1}{3}r$, which at $(t, r) = (5, 4)$ gives $C_{5,4} = 84$ and hence

$$\frac{1}{21} = \frac{4}{84} \leq \delta_{5,4} \leq \frac{1}{20}.$$

We made no search above index 22, so nothing here excludes a $(5, 4)$ broadcast of density below $1/20$.

4. REPRODUCTION AND VERIFICATION

Everything above is determined by the two lines of data

$$\text{basis } \{(10, 0), (4, 2)\}, \quad (x, y) \in S \iff y \equiv 0 \pmod{2}, \quad x \equiv 2y \pmod{10},$$

and a reader who wants only Theorem 1 needs no computer: the twenty receptions of the table are twenty sums of at most three terms each.

The program `verify.py` accompanying this note re-derives from those two lines, in exact integer and rational arithmetic and using only the Python standard library, the quantities printed above: the agreement of the predicate with the span, the index 20 and the transversality of the twenty representatives, the S -periodicity of rec_S , the two table rows entry by entry with their sums and the value $W_5 = 85$, the minimum 4, the density as an exact rational, the shortest nonzero vector, the identity $S = 2T(5, 2)$, the two arguments of Proposition 3, the value $\delta_{4,2} = 1/18$ from the quoted formula with the strict inequality $1/20 < 1/18$, and the constant $C_{5,4} = 84$ with $4/84 = 1/21 < 1/20$. Its 43 checks all pass; the recorded run is `verify.output.txt`. That run also contains checks of statements outside the scope of this note — sublattice censuses at indices 19–22, standard broadcasts, further members of the doubling family, and remarks on published tables and on a further question of Drews, Harris, and Randolph. None of those is claimed here, and nothing above depends on them.

The lower bounds of [2] and [5] are quoted, not reproved — only their arithmetic at the relevant parameters is checked, and the strict inequality of Theorem 1 rests on the theorem of [2]. The program makes no search above index 22 and no literature search.

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